

ADVANCE PROBLEM PROGRAMMING

SLO 1: Classify Paradigms

- 1. Categorize Java, Prolog, SQL, and Python based on paradigm.
- 2. Table showing imperative, declarative, logic, and functional features.
- 3. Explain the hierarchical relationship between paradigms.

Programming paradigms can be viewed hierarchically as different approaches to solving programming problems.

Programming Paradigms

→ Imperative Paradigm → Procedural Programming → Object-Oriented Programming
→ Declarative Paradigm → Functional Programming → Logic Programming → Database Query Languages

Explanation:

Imperative programming focuses on how a task is performed through a sequence of instructions.

Procedural programming organizes imperative instructions into procedures or functions.

Object-oriented programming organizes programs around objects and classes.

Declarative programming focuses on what result is required rather than specifying every execution step.

Functional programming uses functions as the main building blocks of computation.

Logic programming represents knowledge using facts and rules and uses logical reasoning to obtain results.

Language	Main Paradigm	Description
Java	Object-Oriented / Imperative	Uses classes, objects, methods, and statements to control program execution.
Prolog	Logic / Declarative	Programs are based on facts, rules, and logical relationships.
SQL	Declarative	Specifies what data is required rather than how to retrieve it.
Python	Multi-paradigm	Supports object-oriented, imperative, procedural, functional, and some declarative programming features.

Paradigm	Main Idea	Important Features	Example Languages
Imperative	Describes how a task should be performed	Variables, assignments, loops, conditions, sequential statements	Java, Python
Declarative	Describes what result is required	Focuses on the desired result rather than implementation steps	SQL
Logic	Uses facts and rules to derive conclusions	Facts, rules, queries, logical inference	Prolog
Functional	Treats computation as evaluation of functions	Pure functions, immutability, higher-order functions, recursion	Python, Java